

Supporting Information

Supporting Information for “Foundation or Formula? A Simulation-Based Comparison of Google’s TimesFM-3 and Classical Time-Series Models”

Ebrahim Khaled Ebrahim¹ | Somaia Mohamed Ali² | Ahmed El-Kotory³

¹Alexandria University, Egypt | ²Alexandria University, Egypt | ³Alexandria University, Egypt |

Correspondence: Ebrahim Khaled Ebrahim. ebrahimkhaled@alexu.edu.eg

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Summary

This file contains supporting information referred to in the main text as Sections S1 to S7.

S1 | Results under the mean absolute percentage error

Table S1 reports mean MAPE by process, alongside the percentage of series on which it cannot be computed. The D8 row illustrates the problem: every intermittent series contains a zero, so MAPE is undefined for all of them, and the process on which TimesFM-3 achieves its largest and most consistent advantage simply drops out of the comparison. The rankings also move. On D4, MAPE places seasonal naive first (2.360) and TimesFM-3 second (2.801), whereas MASE places AutoARIMA first at every length from $n = 48$. MAPE's asymmetry between over- and under-forecasts is enough to reorder the methods on the same forecasts.

TABLE S1 | Mean MAPE over $h = 1, \dots, 12$ and percentage of series on which MAPE is undefined.

DGP	% undefined	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
D1	0%	2.785	2.737	2.681	2.336	2.501	2.283
D2	0%	3.291	3.341	3.277	2.696	2.971	2.641
D3	0%	9.509	9.278	9.875	10.078	9.393	9.214
D4	0%	2.360	7.471	4.986	2.862	4.538	2.801
D5	0%	2.971	3.583	3.270	2.436	2.916	2.256
D6	0%	2.011	2.379	1.835	1.902	1.925	1.822
D7	0%	7.634	4.631	2.979	2.538	2.767	2.184
D8	100%	—	—	—	—	—	—
D9	0%	1.861	1.769	1.741	1.470	1.599	1.457

Note: On the intermittent process D8 every series contains a zero, so MAPE cannot be computed for it. The best value in each row is in bold.

S2 | Monte Carlo standard errors

Table S2 gives the mean MASE of the six methods of the main design in every cell, with its Monte Carlo standard error.

S3 | Coverage by process

Table S3 gives the empirical coverage of the nominal 80% intervals of the four foundation-model configurations and of AutoARIMA and AutoETS, by process.

S4 | Response-surface regressions

Tables S4 and S5 report the regressions summarised in the section of the main text on randomised parameters and departures from the assumptions.

S5 | Additional results moved from the main text

Table S6 gives the error measures of the six pre-registered methods on the intermittent process D8, and Table S7 compares TimesFM-3 with six Croston-family methods. Table S8 lists the parameter ranges of the randomised design, Figure S1 shows accuracy by forecast horizon, and Figure S2 shows how the relative accuracy of TimesFM-3 and AutoARIMA varies with the drawn parameters. Table S9 gives the mean MASE of every method in every cell of the main design, summarised in Figure 2 of the main text.

S6 | Supplementary diagnostics

Table S10 summarises each method's ratio to the best method of a cell with and without the two cells in which only two seasonal cycles are observed (D4 and D5 at $n = 24$). Table S11 lists the model forms chosen by AutoETS and AutoARIMA on the seasonal processes at $n = 24$ and $n = 48$, and Table S12 gives the 80% interval score by process. Pooling all 180 tests

TABLE S2 | Mean MASE and Monte Carlo standard errors, main design.

DGP	n	Seasonal naive	Theta	AutoETS	AutoARIMA	Combination	TimesFM-3
D1	24	1.646 (0.056)	1.682 (0.055)	1.568 (0.050)	1.485 (0.048)	1.538 (0.048)	1.417 (0.042)
D1	48	1.625 (0.058)	1.586 (0.055)	1.570 (0.054)	1.415 (0.043)	1.478 (0.048)	1.381 (0.039)
D1	96	1.626 (0.054)	1.572 (0.053)	1.564 (0.053)	1.324 (0.039)	1.435 (0.044)	1.307 (0.033)
D1	200	1.615 (0.061)	1.576 (0.057)	1.576 (0.057)	1.266 (0.035)	1.417 (0.046)	1.254 (0.033)
D2	24	1.905 (0.066)	2.003 (0.073)	1.886 (0.069)	1.708 (0.087)	1.788 (0.065)	1.675 (0.062)
D2	48	1.838 (0.068)	1.860 (0.069)	1.834 (0.068)	1.575 (0.053)	1.688 (0.058)	1.518 (0.042)
D2	96	1.810 (0.057)	1.812 (0.057)	1.811 (0.057)	1.517 (0.038)	1.643 (0.045)	1.455 (0.035)
D2	200	1.922 (0.067)	1.919 (0.067)	1.921 (0.067)	1.374 (0.034)	1.654 (0.050)	1.383 (0.033)
D3	24	4.411 (0.213)	4.483 (0.219)	4.892 (0.235)	5.275 (0.256)	4.659 (0.224)	4.656 (0.236)
D3	48	4.252 (0.208)	4.179 (0.206)	4.547 (0.207)	4.679 (0.239)	4.313 (0.203)	4.274 (0.205)
D3	96	4.125 (0.191)	4.135 (0.189)	3.938 (0.179)	3.940 (0.179)	3.871 (0.172)	4.023 (0.177)
D3	200	4.512 (0.210)	4.427 (0.211)	4.275 (0.199)	4.195 (0.201)	4.199 (0.199)	4.297 (0.198)
D4	24	1.029 (0.026)	1.146 (0.110)	3.602 (0.101)	2.284 (0.079)	3.084 (0.072)	1.321 (0.031)
D4	48	1.021 (0.024)	1.944 (0.048)	1.720 (0.032)	1.014 (0.030)	1.395 (0.028)	1.295 (0.027)
D4	96	1.005 (0.024)	2.910 (0.078)	2.407 (0.055)	0.886 (0.020)	1.839 (0.040)	1.124 (0.022)
D4	200	1.053 (0.022)	4.155 (0.117)	1.085 (0.024)	0.886 (0.021)	1.700 (0.043)	1.161 (0.020)
D5	24	1.139 (0.038)	3.154 (0.102)	3.025 (0.093)	1.574 (0.043)	2.474 (0.069)	1.152 (0.038)
D5	48	1.108 (0.032)	0.956 (0.038)	0.741 (0.027)	0.717 (0.024)	0.731 (0.023)	0.815 (0.029)
D5	96	1.076 (0.033)	0.757 (0.025)	0.761 (0.029)	0.732 (0.027)	0.713 (0.027)	0.761 (0.029)
D5	200	1.040 (0.030)	0.759 (0.020)	0.681 (0.024)	0.626 (0.022)	0.645 (0.022)	0.689 (0.024)
D6	24	0.801 (0.025)	1.327 (0.038)	0.766 (0.024)	0.802 (0.025)	0.837 (0.025)	0.807 (0.036)
D6	48	0.947 (0.028)	1.115 (0.031)	0.869 (0.023)	0.925 (0.027)	0.924 (0.025)	0.825 (0.022)
D6	96	1.006 (0.028)	0.963 (0.026)	0.905 (0.023)	0.920 (0.023)	0.916 (0.023)	0.865 (0.021)
D6	200	0.934 (0.023)	0.840 (0.017)	0.825 (0.017)	0.833 (0.017)	0.827 (0.017)	0.844 (0.019)
D7	24	5.417 (0.080)	3.179 (0.058)	1.272 (0.042)	1.083 (0.037)	1.680 (0.048)	1.142 (0.033)
D7	48	4.243 (0.067)	2.262 (0.048)	1.602 (0.061)	1.076 (0.051)	0.970 (0.038)	0.751 (0.022)
D7	96	1.655 (0.042)	0.821 (0.020)	1.305 (0.047)	1.329 (0.048)	1.097 (0.036)	0.959 (0.032)
D7	200	0.917 (0.026)	1.301 (0.031)	0.966 (0.028)	0.994 (0.030)	0.954 (0.027)	0.933 (0.025)
D8	24	1.019 (0.054)	0.999 (0.034)	1.047 (0.058)	0.996 (0.032)	1.010 (0.039)	0.780 (0.031)
D8	48	1.066 (0.066)	0.984 (0.036)	0.988 (0.033)	0.962 (0.024)	0.974 (0.028)	0.708 (0.030)
D8	96	0.932 (0.047)	0.923 (0.017)	0.926 (0.015)	0.924 (0.015)	0.923 (0.015)	0.683 (0.022)
D8	200	1.005 (0.052)	0.943 (0.017)	0.939 (0.015)	0.939 (0.015)	0.940 (0.015)	0.694 (0.023)
D9	24	1.547 (0.061)	1.518 (0.057)	1.454 (0.057)	1.372 (0.057)	1.404 (0.054)	1.321 (0.049)
D9	48	1.588 (0.056)	1.486 (0.054)	1.476 (0.053)	1.244 (0.041)	1.353 (0.046)	1.239 (0.039)
D9	96	1.430 (0.051)	1.342 (0.048)	1.333 (0.048)	1.098 (0.034)	1.213 (0.040)	1.104 (0.034)
D9	200	1.577 (0.056)	1.503 (0.053)	1.504 (0.053)	1.190 (0.032)	1.341 (0.042)	1.200 (0.035)

Note: Mean MASE over $h = 1, \dots, 12$ with its Monte Carlo standard error in parentheses (standard deviation over the 200 replications divided by $\sqrt{200}$).

TABLE S3 | Coverage of 80% prediction intervals by process.

Process	TimesFM-3	TimesFM-3 (eval.)	TimesFM-2.5	Chronos-Bolt	AutoARIMA	AutoETS
D1	0.817	0.819	0.787	0.814	0.777	0.852
D2	0.817	0.818	0.782	0.824	0.766	0.874
D3	0.704	0.707	0.703	0.721	0.706	0.618
D4	0.846	0.856	0.810	0.769	0.756	0.661
D5	0.818	0.830	0.799	0.833	0.724	0.686
D6	0.859	0.873	0.913	0.921	0.986	0.987
D7	0.845	0.848	0.894	0.968	0.833	0.765
D8	0.818	0.869	0.835	0.846	0.836	0.843
D9	0.807	0.809	0.783	0.810	0.760	0.825

Note: Empirical coverage of the nominal 80% interval $[q_{0.1}, q_{0.9}]$, averaged over lengths and replications; series-cluster bootstrap Monte Carlo SE at most 0.017.

of the $h = 1$ –12 slice into one Benjamini–Hochberg family gives 134 significant comparisons, 115 won by TimesFM-3 (13–9 against AutoARIMA); pooling all 540 tests gives 132 and 113 (12–9), against 132 and 113 (11–9) with the pre-registered families.

S7 | A longer horizon

The main design forecasts 12 steps. To see whether its conclusions depend on that choice, the nine processes were regenerated with the same seeds at length $n + 48$ and forecast 48 steps ahead by the six pre-registered methods, TimesFM-2.5

TABLE S4 | Response-surface regressions of the robustness study, processes D1–D4.

Process	Term	Coefficient	95% interval	<i>p</i>	<i>R</i> ²
D1	version: est period	0.000	[-0.002, 0.002]	0.945	0.003
D1	version: heavy tail	-0.022	[-0.048, 0.004]	0.091	0.003
D1	version: outliers	-0.010	[-0.031, 0.010]	0.335	0.003
D1	$\log_2 n$	0.004	[-0.010, 0.019]	0.552	0.003
D1	parameter phi	-0.013	[-0.031, 0.005]	0.161	0.003
D1	parameter sigma	0.006	[-0.010, 0.021]	0.476	0.003
D2	version: est period	-0.000	[-0.001, 0.001]	0.795	0.012
D2	version: heavy tail	-0.016	[-0.041, 0.009]	0.207	0.012
D2	version: outliers	0.004	[-0.014, 0.021]	0.661	0.012
D2	$\log_2 n$	-0.002	[-0.017, 0.013]	0.828	0.012
D2	parameter phi	-0.031	[-0.050, -0.012]	0.002	0.012
D2	parameter sigma	0.010	[-0.007, 0.027]	0.232	0.012
D2	parameter theta	0.005	[-0.012, 0.022]	0.567	0.012
D3	version: est period	0.000	[-0.004, 0.005]	0.876	0.006
D3	version: heavy tail	0.020	[-0.036, 0.075]	0.487	0.006
D3	version: outliers	-0.004	[-0.039, 0.032]	0.843	0.006
D3	$\log_2 n$	0.014	[-0.021, 0.049]	0.422	0.006
D3	parameter phi	-0.046	[-0.088, -0.004]	0.030	0.006
D3	parameter sigma	-0.004	[-0.043, 0.035]	0.844	0.006
D3	parameter theta	0.017	[-0.022, 0.056]	0.396	0.006
D3	parameter drift	-0.010	[-0.049, 0.029]	0.608	0.006
D4	version: est period	-0.004	[-0.009, 0.001]	0.141	0.255
D4	version: heavy tail	0.061	[0.028, 0.095]	0.000	0.255
D4	version: outliers	-0.219	[-0.260, -0.178]	0.000	0.255
D4	$\log_2 n$	0.258	[0.229, 0.287]	0.000	0.255
D4	parameter phi	-0.055	[-0.089, -0.021]	0.002	0.255
D4	parameter sigma	0.047	[0.012, 0.082]	0.009	0.255
D4	parameter Phi	0.110	[0.075, 0.145]	0.000	0.255
D4	parameter amp	-0.038	[-0.071, -0.006]	0.020	0.255

Note: One ordinary least-squares regression per process of the per-series $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA on $\log_2 n$ (not standardised), the version (reference: random parameters with Gaussian innovations) and the drawn parameters, each standardised to mean 0 and standard deviation 1. Standard errors are clustered by parameter draw, because the four versions of a draw share its parameters and normal draws; 3 200 series per process (3 188 for D8, where series with a zero MASE denominator are excluded). Positive coefficients move the ratio against TimesFM-3.

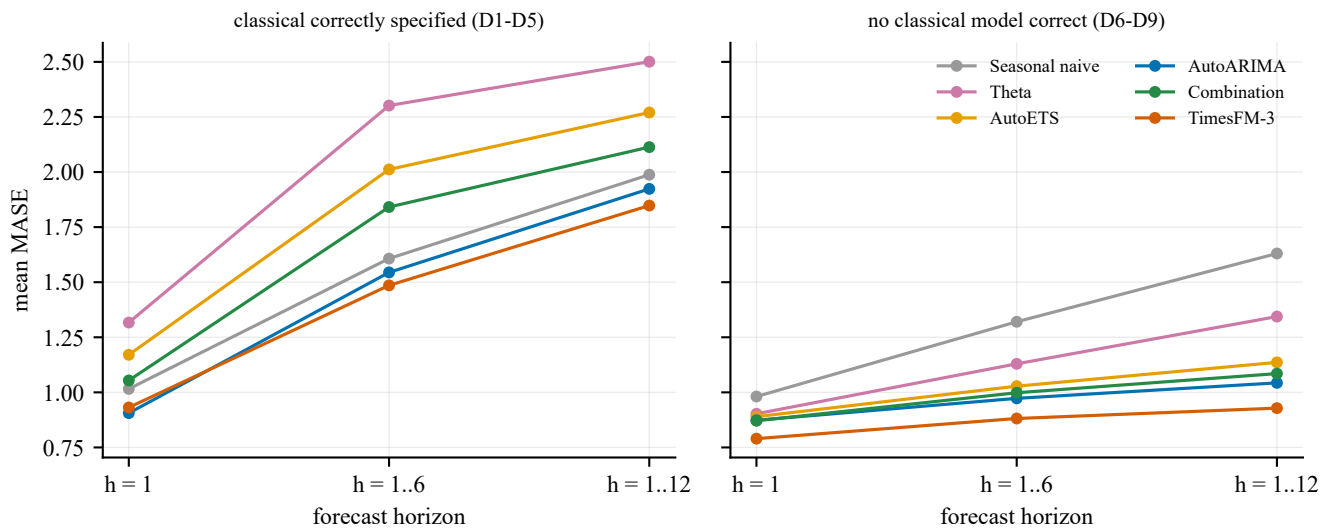


FIGURE S1 | Mean MASE by forecast horizon, averaged within each regime. Left: D1–D5, for which a classical family is correctly specified. Right: D6–D9, for which none is, or only in the conditional mean (D9).

TABLE S5 | Response-surface regressions of the robustness study, processes D5–D9.

Process	Term	Coefficient	95% interval	<i>p</i>	<i>R</i> ²
D5	version: est period	-0.086	[-0.111, -0.061]	0.000	0.090
D5	version: heavy tail	-0.002	[-0.039, 0.036]	0.934	0.090
D5	version: outliers	-0.151	[-0.191, -0.111]	0.000	0.090
D5	$\log_2 n$	0.147	[0.119, 0.174]	0.000	0.090
D5	parameter sigma	0.032	[-0.001, 0.065]	0.056	0.090
D5	parameter amp	-0.021	[-0.054, 0.012]	0.214	0.090
D5	parameter alpha	-0.017	[-0.050, 0.016]	0.303	0.090
D5	parameter beta	-0.008	[-0.042, 0.026]	0.659	0.090
D5	parameter gamma	0.031	[-0.003, 0.064]	0.075	0.090
D5	parameter trend	0.010	[-0.022, 0.042]	0.522	0.090
D6	version: est period	-0.002	[-0.004, 0.000]	0.066	0.022
D6	version: heavy tail	0.018	[-0.022, 0.058]	0.384	0.022
D6	version: outliers	-0.023	[-0.053, 0.008]	0.140	0.022
D6	$\log_2 n$	-0.046	[-0.080, -0.012]	0.007	0.022
D6	parameter sigma	-0.029	[-0.070, 0.011]	0.154	0.022
D6	parameter jump sd	-0.038	[-0.071, -0.004]	0.028	0.022
D6	parameter break pos	-0.047	[-0.085, -0.009]	0.016	0.022
D6	parameter sigma level	0.029	[-0.006, 0.064]	0.100	0.022
D6	parameter jump sign	-0.011	[-0.047, 0.025]	0.549	0.022
D7	version: est period	-0.002	[-0.011, 0.008]	0.741	0.022
D7	version: heavy tail	-0.031	[-0.081, 0.018]	0.216	0.022
D7	version: outliers	-0.045	[-0.094, 0.005]	0.075	0.022
D7	$\log_2 n$	0.034	[-0.007, 0.075]	0.103	0.022
D7	parameter sigma	-0.048	[-0.100, 0.004]	0.070	0.022
D7	parameter r	0.068	[0.017, 0.120]	0.009	0.022
D7	parameter infl pos	-0.081	[-0.136, -0.027]	0.003	0.022
D7	parameter K	0.011	[-0.039, 0.061]	0.678	0.022
D8	version: est period	-0.001	[-0.003, 0.001]	0.355	0.203
D8	version: heavy tail	-0.047	[-0.078, -0.016]	0.003	0.203
D8	version: outliers	-0.025	[-0.043, -0.007]	0.007	0.203
D8	$\log_2 n$	-0.096	[-0.162, -0.030]	0.004	0.203
D8	parameter p	0.509	[0.422, 0.596]	0.000	0.203
D8	parameter lam	0.008	[-0.064, 0.080]	0.836	0.203
D9	version: est period	-0.002	[-0.003, -0.000]	0.045	0.000
D9	version: heavy tail	-0.007	[-0.032, 0.018]	0.592	0.000
D9	version: outliers	-0.013	[-0.032, 0.007]	0.196	0.000
D9	$\log_2 n$	-0.003	[-0.018, 0.012]	0.719	0.000
D9	parameter phi	0.002	[-0.017, 0.021]	0.814	0.000
D9	parameter a	-0.001	[-0.019, 0.017]	0.923	0.000
D9	parameter b	-0.001	[-0.018, 0.016]	0.927	0.000

Note: One ordinary least-squares regression per process of the per-series $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA on $\log_2 n$ (not standardised), the version (reference: random parameters with Gaussian innovations) and the drawn parameters, each standardised to mean 0 and standard deviation 1. Standard errors are clustered by parameter draw, because the four versions of a draw share its parameters and normal draws; 3 200 series per process (3 188 for D8, where series with a zero MASE denominator are excluded). Positive coefficients move the ratio against TimesFM-3.

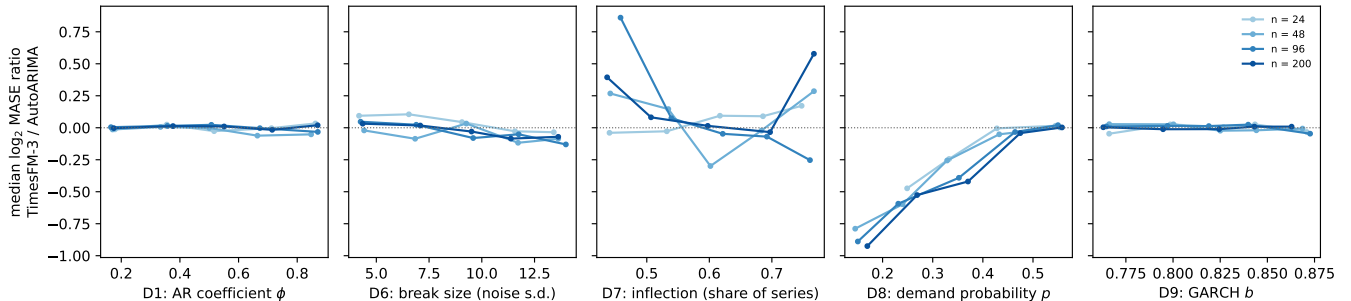
**FIGURE S2** | Median $\log_2$ ratio of the MASE of TimesFM-3 to that of AutoARIMA, by quintile of the drawn parameter, randomised-parameter design with Gaussian innovations. Values below zero favour TimesFM-3. For D8 the ratio compares median-type forecasts and is shown for completeness only.

TABLE S6 | Accuracy on the intermittent process D8 under four error measures.

Metric	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
MASE	24	1.019	0.999	1.047	0.996	1.010	0.780
	48	1.066	0.984	0.988	0.962	0.974	0.708
	96	0.932	0.923	0.926	0.924	0.923	0.683
	200	1.005	0.943	0.939	0.939	0.940	0.694
RMSSE	24	0.940	0.791	0.792	0.765	0.775	0.806
	48	0.937	0.746	0.726	0.718	0.724	0.774
	96	0.875	0.707	0.697	0.698	0.697	0.775
	200	0.923	0.718	0.706	0.706	0.706	0.797
sMAPE	24	86.696	172.741	169.988	165.908	170.489	191.774
	48	84.690	172.522	170.746	170.622	171.026	196.183
	96	84.171	170.801	169.574	170.065	169.949	197.610
	200	85.990	170.003	169.189	169.315	169.236	198.556
Pinball	24	0.634	0.395	0.402	0.380	0.388	0.325
	48	0.653	0.381	0.368	0.360	0.365	0.307
	96	0.614	0.348	0.341	0.341	0.341	0.288
	200	0.628	0.354	0.344	0.345	0.345	0.291

Note: Means over $h = 1, \dots, 12$ and 200 replications per length. The pinball loss is averaged over the nine deciles. Lower is better; the best value in each row is in bold.

TABLE S7 | Intermittent-demand baselines on the process D8 (post hoc analysis).

Metric	n	Croston	Croston-opt	Croston-SBA	ADIDA	IMAPA	TSB	AutoARIMA	TimesFM-3
MASE	24	1.042	1.024	1.025	0.972	0.972	0.971	0.996	0.780
	48	0.990	0.992	0.974	0.947	0.954	0.966	0.962	0.708
	96	0.935	0.948	0.921	0.926	0.923	0.916	0.924	0.683
	200	0.947	0.948	0.934	0.941	0.942	0.948	0.939	0.694
RMSSE	24	0.769	0.758	0.763	0.753	0.757	0.764	0.765	0.806
	48	0.722	0.726	0.718	0.717	0.722	0.733	0.718	0.774
	96	0.696	0.702	0.694	0.700	0.701	0.716	0.698	0.775
	200	0.709	0.710	0.707	0.709	0.712	0.730	0.706	0.797

Note: Mean MASE and RMSSE over $h = 1, \dots, 12$ and 200 replications per length. Croston-opt is Croston's method with optimised smoothing parameters and Croston-SBA the Syntetos–Boylan approximation. Lower is better; the best value in each row is in bold.

TABLE S8 | Parameter ranges of the randomised-parameter design.

ID	Parameters, each drawn uniformly on the stated interval
D1	$\phi \in [0.10, 0.95], \sigma \in [1, 5]$
D2	$\phi \in [0.10, 0.90], \theta \in [-0.6, 0.8], \sigma \in [1, 5]$
D3	$\phi \in [-0.5, 0.8], \theta \in [-0.5, 0.6], \text{drift} \in [0, 0.6], \sigma \in [1, 5]$
D4	$\phi \in [0.1, 0.8], \Phi \in [0.1, 0.8], \text{seasonal amplitude} \in [5, 20], \sigma \in [1, 5]$
D5	$\alpha \in [0.05, 0.5], \beta \in [0.001, 0.03], \gamma \in [0.05, 0.3], \text{initial trend} \in [0, 0.5], \text{amplitude} \in [5, 20], \sigma \in [1, 5]$
D6	$\text{jump} \in [3, 15] \sigma \text{ with random sign, break at } [0.50, 0.85] n, \text{level noise} \in [0.2, 1], \sigma \in [1, 5]$
D7	$r \in [0.04, 0.15], \text{inflection at } [0.4, 0.8](n + H), K \in [100, 250], \sigma \in [1, 6]$
D8	demand probability $p \in [0.1, 0.6], \text{size } 1 + \text{Pois}(\lambda) \text{ with } \lambda \in [1, 8]$
D9	$\phi \in [0.1, 0.9], a \in [0.05, 0.15], b \in [0.75, 0.90]; \text{if } a + b \geq 0.98, b \text{ is set to } 0.97 - a$

Note: Parameters not listed are as in Table 2 of the main text. The intervals contain the fixed values of the main design. The rule for D9 keeps the conditional variance stationary and puts a small point mass at $b = 0.97 - a$. Seeds are disjoint from those of the main design; the four versions share the parameter draws and, except for the heavy-tailed D8 sizes, the underlying normal draws.

and Chronos-2 (post hoc). Because the inflection of D7 sits at $0.6(n + H)$, the longer horizon moves it into the context for $n = 200$: the series has reached its plateau before the forecast origin. Table S14 and Figure S3 summarise the results.

TABLE S9 | Mean MASE over $h = 1, \dots, 12$ across 200 replications per cell.

DGP	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	TimesFM-3
D1 (ARIMA)	24	1.646	1.682	1.568	1.485	1.538	1.417
	48	1.625	1.586	1.570	1.415	1.478	1.381
	96	1.626	1.572	1.564	1.324	1.435	1.307
	200	1.615	1.576	1.576	1.266	1.417	1.254
D2 (ARIMA)	24	1.905	2.003	1.886	1.708	1.788	1.675
	48	1.838	1.860	1.834	1.575	1.688	1.518
	96	1.810	1.812	1.811	1.517	1.643	1.455
	200	1.922	1.919	1.921	1.374	1.654	1.383
D3 (ARIMA)	24	4.411	4.483	4.892	5.275	4.659	4.656
	48	4.252	4.179	4.547	4.679	4.313	4.274
	96	4.125	4.135	3.938	3.940	3.871	4.023
	200	4.512	4.427	4.275	4.195	4.199	4.297
D4 (ARIMA)	24	1.029	4.146	3.602	2.284	3.084	1.321
	48	1.021	1.944	1.720	1.014	1.395	1.295
	96	1.005	2.910	2.407	0.886	1.839	1.124
	200	1.053	4.155	1.085	0.886	1.700	1.161
D5 (ETS)	24	1.139	3.154	3.025	1.574	2.474	1.152
	48	1.108	0.956	0.741	0.717	0.731	0.815
	96	1.076	0.757	0.761	0.732	0.713	0.761
	200	1.040	0.759	0.681	0.626	0.645	0.689
D6 (none)	24	0.801	1.327	0.766	0.802	0.837	0.807
	48	0.947	1.115	0.869	0.925	0.924	0.825
	96	1.006	0.963	0.905	0.920	0.916	0.865
	200	0.934	0.840	0.825	0.833	0.827	0.844
D7 (none)	24	5.417	3.179	1.272	1.083	1.680	1.142
	48	4.243	2.262	1.602	1.076	0.970	0.751
	96	1.655	0.821	1.305	1.329	1.097	0.959
	200	0.917	1.301	0.966	0.994	0.954	0.933
D8 (none)	24	1.019	0.999	1.047	0.996	1.010	0.780
	48	1.066	0.984	0.988	0.962	0.974	0.708
	96	0.932	0.923	0.926	0.924	0.923	0.683
	200	1.005	0.943	0.939	0.939	0.940	0.694
D9 (ARIMA in mean)	24	1.547	1.518	1.454	1.372	1.404	1.321
	48	1.588	1.486	1.476	1.244	1.353	1.239
	96	1.430	1.342	1.333	1.098	1.213	1.104
	200	1.577	1.503	1.504	1.190	1.341	1.200

Note: Lower is better; the best method in each row is in bold. The family in parentheses after each process label is the one correctly specified for it by construction.

TABLE S10 | Ratio of each method's mean MASE to the best in the same cell, with and without the two-cycle cells.

Method	Median	Geom. mean	90th pct.	Worst	Worst excl.	Cell of worst excl.
Seasonal naive	1.200	1.331	1.60	5.65	5.65	D7, $n = 48$
Theta	1.251	1.458	2.97	4.69	4.69	D4, $n = 200$
AutoETS	1.194	1.296	1.91	3.50	2.72	D4, $n = 96$
AutoARIMA	1.032	1.118	1.37	2.22	1.62	D7, $n = 96$
Combination	1.108	1.228	1.74	3.00	2.08	D4, $n = 96$
TimesFM-3	1.008	1.050	1.22	1.31	1.31	D4, $n = 200$

Note: Main design, 36 cells, $h = 1, \dots, 12$. "Excl." leaves out D4 and D5 at $n = 24$, where only two seasonal cycles are observed; the 90th percentile is taken over the 36 cells.

TABLE S11 | Model forms chosen by AutoETS and AutoARIMA on the seasonal processes, given $m = 12$.

Process	n	ETS seas. (%)	ARIMA seas. (%)	Most frequent ETS forms	Most frequent ARIMA orders
D4	24	8.5	0.0	ETS(A,N,N) (100), ETS(M,N,N) (83), ETS(A,N,A) (8)	(2,0,1)(0,0,0) (52), (3,0,1)(0,0,0) (44)
D4	48	98.0	100.0	ETS(A,N,A) (67), ETS(M,N,M) (60), ETS(M,N,A) (42)	(1,0,0)(1,1,0) (52), (1,0,0)(0,1,1) (26)
D5	24	0.0	0.0	ETS(A,N,N) (117), ETS(M,N,N) (83)	(2,0,1)(0,0,0) (72), (4,0,0)(0,0,0) (52)
D5	48	100.0	100.0	ETS(M,N,M) (54), ETS(A,A,A) (40), ETS(M,A,A) (26)	(0,1,1)(0,1,1) (48), (0,0,0)(0,1,1) (35)

Note: 200 replications per row, the series of the main design; counts out of 200 in parentheses. A form is seasonal when it has a seasonal component (ETS) or a non-zero seasonal order (ARIMA).

TABLE S12 | Scaled 80% interval score by process, averaged over the four lengths.

Process	Seasonal naive	Theta	AutoETS	AutoARIMA	Combination	TimesFM-3
D1	8.49	7.80	7.67	6.36	6.97	6.05
D2	9.05	9.28	9.09	6.98	8.00	6.86
D3	23.33	23.31	22.12	22.03	21.01	21.24
D4	4.58	16.77	11.04	5.65	9.65	5.69
D5	4.64	7.22	6.51	4.29	5.59	3.98
D6	9.35	7.13	7.21	7.53	7.28	4.55
D7	14.14	8.59	5.91	5.17	5.51	4.55
D8	10.11	4.60	4.50	4.40	4.47	3.54
D9	8.56	7.22	7.07	5.82	6.44	5.67

Note: Interval score of $[q_{0.1}, q_{0.9}]$ with $\alpha = 0.2$ (width plus $2/\alpha$ times the distance by which the outcome falls outside the interval), averaged over the horizon and scaled by the MASE denominator, as for MSIS; lower is better, the best in bold. TimesFM-3 has the lowest score in 22 of the 36 (process, length) cells.

TABLE S13 | Classical methods in forecast (R) and StatsForecast (Python) on the seasonal processes: mean MASE over $h = 1, \dots, 12$.

Process	n	R ARIMA	SF ARIMA	R ETS	SF ETS	R Theta	SF Theta	R seas. (%)
D4	24	2.866	2.284	3.949	3.602	4.140	4.146	0 / 13
D4	48	1.008	1.014	1.765	1.720	1.949	1.944	100 / 100
D4	96	0.891	0.886	1.044	2.407	2.896	2.910	100 / 100
D4	200	0.894	0.886	1.056	1.085	4.085	4.155	100 / 100
D5	24	1.695	1.574	3.304	3.025	3.147	3.154	0 / 0
D5	48	0.717	0.717	0.739	0.741	0.947	0.956	100 / 100
D5	96	0.747	0.732	0.719	0.761	0.759	0.757	100 / 100
D5	200	0.631	0.626	0.644	0.681	0.760	0.759	100 / 100

Note: The same 200 series per row. R: `auto.arima`, `ets` and `thetaf` of forecast 9.0.2 at their defaults, with frequency 12; SF: StatsForecast 2.1.1 as in the main text. R seas.: share of series given a seasonal form by `auto.arima` / `ets`. No R fit failed. On D4 at $n = 96$ StatsForecast's AutoETS settles on a near-zero seasonal smoothing parameter with a much lower likelihood than R's fit of the same form.

TABLE S14 | Longer horizon ($H = 48$): worst ratio to the cell-best and 80% coverage by block of steps.

Method	Worst ratio, D7 excluded			Coverage 80%			D7, $n = 200$ MASE 1–48
	1–12	13–24	25–48	1–12	13–24	25–48	
TimesFM-3	1.31	1.39	1.80	0.788	0.774	0.760	8.37
TimesFM-2.5	1.42	1.59	1.77	0.781	0.734	0.707	12.22
Chronos-2	2.94	2.71	3.33	0.787	0.762	0.761	5.30
AutoARIMA	2.11	1.60	1.41	0.760	0.739	0.720	2.51
AutoETS	3.49	2.17	1.67	0.779	0.790	0.784	2.35
Theta	4.65	3.66	6.27	0.731	0.757	0.768	2.73
Combination	2.92	2.00	2.17	0.788	0.812	0.814	2.22
Seasonal naive	1.66	1.88	1.87	0.805	0.780	0.753	1.11

Note: The nine processes regenerated with the same seeds at length $n + 48$, 200 replications per cell; ratios to the best of the eight methods in each of the 32 (process, length) cells without D7. Coverage is averaged over all 36 cells. The last column is the mean MASE over the 48 steps on D7 at $n = 200$, where the process has reached its plateau before the forecast origin.

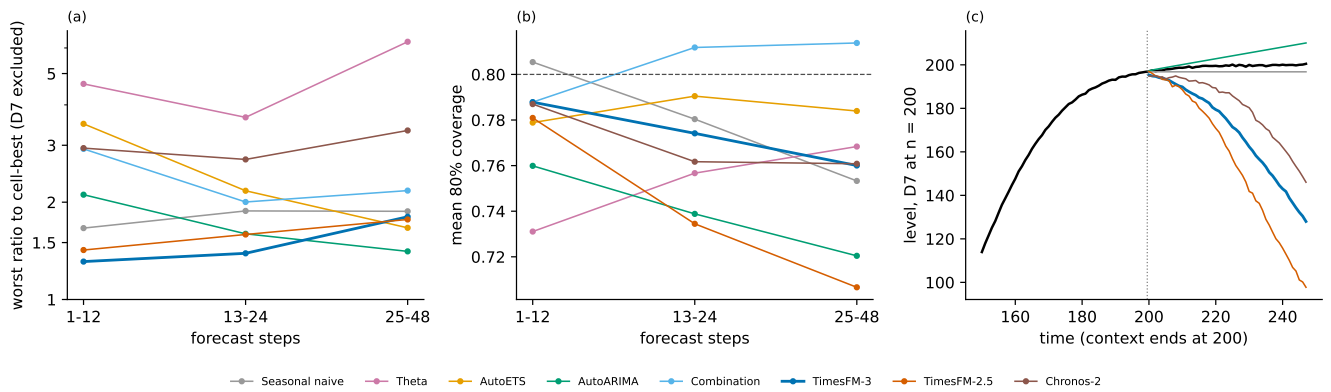


FIGURE S3 | Longer horizon ($H = 48$). (a) Worst ratio to the cell-best method by block of forecast steps, D7 excluded. (b) Mean coverage of the nominal 80% intervals by block; the dashed line is the nominal level. (c) D7 at $n = 200$: mean of the 200 series (black) and mean point forecasts over the 48 steps.